Annexure-3

AMITY UNIVERSITY
MADHYA PRADESH
Established vide Government of Madhya Pradesh Act No. 27 of 2010

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
Course Handout
Course : INTERNET OF THINGS (IOT) LAB
Course Code : CSE623, Crédits **:** 01, **Session :**2023-27(Even Sem.), **Class :** B. Tech. 3rd Year

Introduction: The objective of the course is to Introduction of IOT, Understand IOT Market perspective, Data and Knowledge Management and use of Devices in IOT Technology, Understand State of the Art – IOT Architecture. Real World IOT Design Constraints, Industrial Automation and Commercial Building Automation in IOT.

Course Outcomes: At the end of the course, students will be able to:

CO1: Recall the fundamental concept of the Python in Eclipse Study, Install and Demonstrate basic operations.

CO2: Describe and Implement the different types of Arduino, Install IDE and perform basic LED programs.

CO3: Discuss and Demonstrate RFID, NFC and MQTT.

CO4: Compare Arduino with Raspberry Pi and Demonstrate Raspberry Pi basic LED programs and Zigbee Protocol.

Syllabus

1. Study and Install Python in Eclipse and WAP for data types in python. (3 Hours)
2. Write a Program for arithmetic operation in Python. (3 Hours)
3. Write a Program for looping statement in Python. (3 Hours)
4. Study and Install IDE of Arduino and different types of Arduino. (3 Hours)
5. Write program using Arduino IDE for Blink LED. (3 Hours)
6. Write Program for RGB LED using Arduino. (3 Hours)
7. Study the Temperature sensor and Write Program for monitor temperature using Arduino (2 Hours)
8. Study and Implement RFID, NFC using Arduino. (2 Hours)
9. Study and implement MQTT protocol using Arduino. (2Hours)
10. Study and Configure Raspberry Pi. (2 Hours)
11. WAP for LED blink using Raspberry Pi. (2 Hours)
12. Study and Implement Zigbee Protocol using Arduino / Raspberry Pi. (2Hours)

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Examination Scheme:

IA			EE			
A	**PR**	**Practical Based Test**	**Major Experiment**	**Minor Experiment**	**LR**	**Viva**
5	10	15	35	15	10	10

Note: IA – Internal Assessment, EE - External Exam, A - Attendance, PR- Performance, LR – Lab Record, V – Viva.

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Lab Plan

Lab Session	Program Name	Mapped CO	Mode of Assessing CO
1	Study and Install Python in Eclipse and WAP for data types in python.	**CO1**	Viva/Quiz & End Sem Practical Exam
2	Write a Program for arithmetic operation in Python.	**CO1**	Viva/Quiz & End Sem Practical Exam
3	Write a Program for looping statement in Python.	**CO1**	Viva/Quiz & End Sem Practical Exam
4	Study and Install IDE of Arduino and different types of Arduino.	**CO2**	Viva/Quiz & End Sem Practical Exam
5	Write program using Arduino IDE for Blink LED.	**CO2**	Viva/Quiz & End Sem Practical Exam
6	Write Program for RGB LED using Arduino.	**CO2**	Viva/Quiz & End Sem Practical Exam
7	Study the Temperature sensor and Write Program for monitor temperature using Arduino	**CO2**	Viva/Quiz & End Sem Practical Exam
8	Explain working and WAP to implement ultrasonic sensor using Arduino.	**CO2**	Viva/Quiz & End Sem Practical Exam
9	Study and Implement RFID/ NFC using Arduino.	**CO3**	Viva/Quiz & End Sem Practical Exam
10	Study and implement MQTT protocol using Arduino.	**CO3**	Viva/Quiz & End Sem Practical Exam
11	Study and Implement Zigbee Protocol using Arduino.	**CO3**	Viva/Quiz & End Sem Practical Exam
12	Study and Configure Raspberry Pi.	**CO4**	Viva/Quiz & End Sem Practical Exam
13	WAP for LED blink using Raspberry Pi.	**CO4**	Viva/Quiz & End Sem Practical Exam
14	Explain working and WAP to implement ultrasonic sensor using Raspberry Pi..	**CO4**	Viva/Quiz & End Sem Practical Exam
15	Study and Implement Zigbee Protocol using Raspberry Pi.	**CO4**	Viva/Quiz & End Sem Practical Exam

16. Course Articulation Matrix (Mapping of COs with POs)

CO	CORRELATION WITH PROGRAMME OUTCOMES												CORRELATION WITH PROGRAMME SPECIFIC OUTCOMES		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	1	1	1	2				1		1	1			
CO2	3	2	1	2	2				1		1	1			
CO3	3	2	2	2	2				1		1	1			
CO4	3	2	2	2	2				1		1	1			

1: strongly related, 2: moderately related and 3: weakly related